

1750 — 2024

Global CO₂ Emissions: Trends, Regional Shifts, and Per Capita Inequalities

A data-driven overview of how global carbon dioxide emissions have evolved from the Industrial Revolution to 2024, highlighting regional shifts, per capita disparities, and the latest annual changes.

Data: Global Carbon Budget 2025 · NASA · Our World in Data

429

parts per million

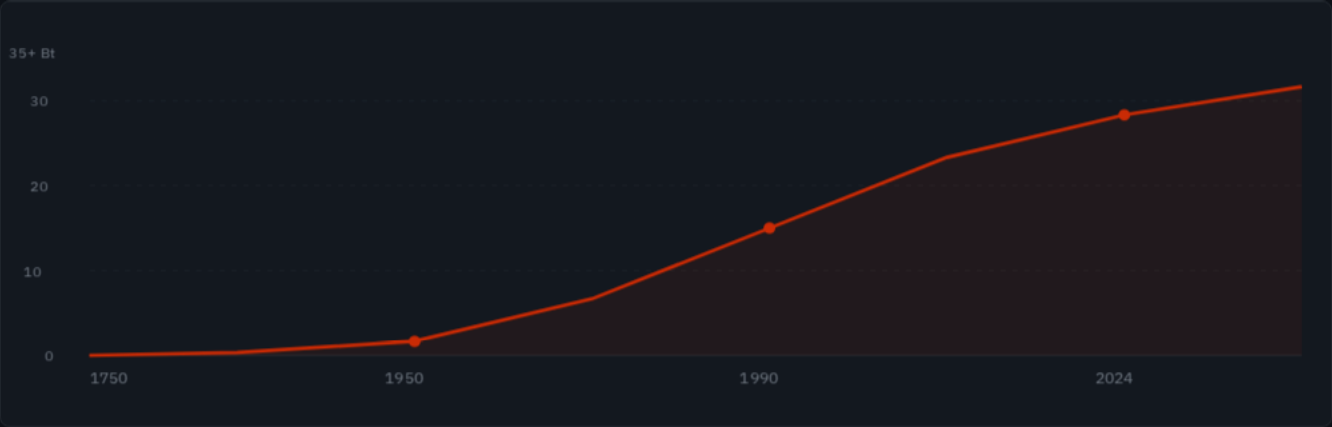
NOAA Mauna Loa Observatory, February 2026. More than 50% above the pre-industrial baseline of ~280 ppm — a change that took thousands of years at the end of the last ice age, now compressed into roughly two centuries.

- ~280 ppm
Pre-industrial baseline
- 315 ppm
Mauna Loa records begin (1958)
- 429 ppm
February 2026 — >50% increase
- Primary driver
Burning of coal, oil, and natural gas



Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes

Before the Industrial Revolution, CO₂ emissions were negligible. Growth remained slow until the mid-20th century, then accelerated dramatically.



1950

6 Bt

Global CO₂ emissions

1990

20+ Bt

Nearly quadrupled

2024

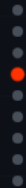
35+ Bt

Still rising

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

China is the world's largest emitter by a significant margin. The US peaked near 6 billion tonnes around 2005 and has since declined to roughly 5 billion tonnes. India has risen rapidly to approximately 3 billion tonnes.

COUNTRY / REGION	ANNUAL CO ₂ (BILLION T)	SHARE OF GLOBAL	TREND
China	~12.5	~27%	Slight increase
United States	~5.0	~14%	Declining
India	~3.0	~8%	Rising
EU-27	~2.7	~7%	Declining
Germany	~0.6	~1.7%	Declining
Brazil	~0.5	~1.4%	Roughly stable
United Kingdom	~0.3	~0.9%	Declining



Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third

1900

>90%

Europe and the United States accounted for more than 90% of global CO₂ emissions.

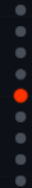
By 1950, they still represented over 85%.

2024

~50%

Asia — home to nearly 60% of the world's population — produces roughly half of all emissions.

US + Europe combined: <33%. Africa & South America: 3–4% each.



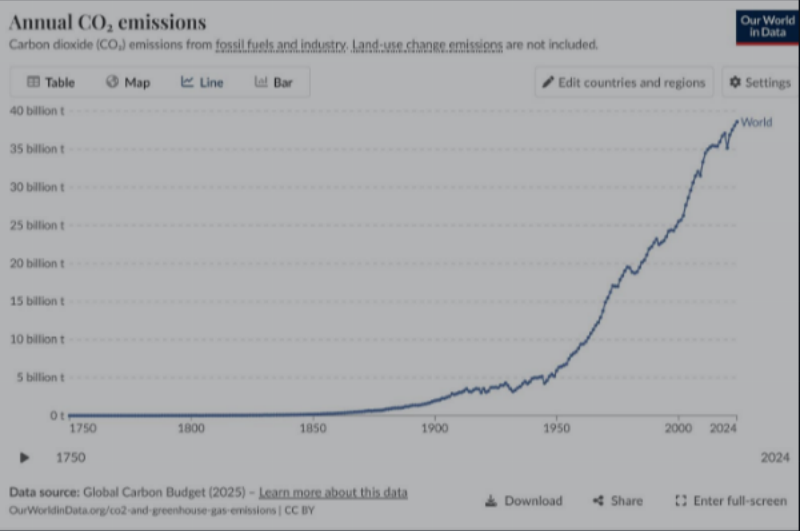
Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity

86 / 88
Per capita CO₂ emissions reveal stark global inequalities. Among large, populous countries, the US, Australia, and Canada have per capita emissions roughly three times the global average.



2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

The 2024 map of annual percentage change in CO₂ emissions shows a mixed picture across the globe.



Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can have very different per capita emissions depending on their reliance on nuclear, renewables, or fossil fuels.

France / UK

Lower per capita CO₂ (~4 t/person)

Nuclear & Renewables	High share
Fossil fuels	Lower share
Per capita CO ₂	~4 t

Germany / Netherlands

Higher per capita CO₂

Nuclear & Renewables	Lower share
Fossil fuels	Higher share
Per capita CO ₂	Higher



Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is Shifting

Synthesizing the data for policy analysts and climate negotiators.

01 Global emissions exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth rates.

02 China alone accounts for over 25% of global emissions (~12.5 Bt), more than double the United States (~5 Bt).

